

HALTEN FYR, Norway

WMO: 012400

Lat: 64.173N

Lon: 9.406E

Elev: 52

StdP: 14.67

Time Zone: 1.00 (EUC)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	20.5	23.9	12.0	10.1	22.4	14.4	11.5	25.1	47.0	33.9	44.6	36.0	20.4	110	0.903

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.1	65.7	59.8	63.3	58.3	61.5	57.0	60.7	64.6	59.1	62.6	57.6	60.8	12.3	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.6	73.7	62.8	57.2	69.9	61.5	55.8	66.4	59.9	26.9	64.9	25.9	62.7	24.9	61.0	65.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 44.9	40.0	35.9	19.2	70.7	5.7	2.8	15.2	72.7	11.9	74.3	8.7	75.8	4.6	77.8		
(5)			16.9	62.4	5.3	2.5	13.1	64.2	10.0	65.7	7.1	67.1	3.2	69.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.3	36.0	35.9	37.2	41.5	46.4	51.0	55.8	56.9	53.9	47.3	42.2	38.4
	DBStd	8.75	5.29	5.83	4.83	3.92	4.40	3.63	3.93	3.23	3.47	4.49	5.42	5.62
	HDD50	2347	434	394	395	257	128	30	1	1	7	106	235	359
	HDD65	7200	899	814	860	705	575	420	286	251	332	550	684	824
	CDD50	625	0	0	0	2	18	59	182	215	125	22	2	0
	CDD65	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH74	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	17.7	23.6	21.3	20.0	15.7	13.7	13.4	12.5	12.7	16.1	19.9	20.7	23.2
Precipitation	PrecAvg	35.3	3.8	3.4	3.2	2.0	1.8	2.3	2.1	2.6	3.5	3.6	3.5	3.8
	PrecMax	41.3	6.3	6.7	7.7	4.5	2.7	4.9	4.4	5.4	8.4	5.9	6.3	7.4
	PrecMin	27.6	0.2	0.5	1.1	0.3	0.7	0.6	0.7	0.8	1.3	0.6	0.2	1.7
	PrecStd	4.2	1.7	1.5	1.6	1.0	0.6	0.9	0.9	1.1	1.9	1.3	2.0	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.9	45.9	47.4	55.7	61.8	63.9	71.2	68.0	62.8	58.3	53.1	49.1
		MCWB	44.1	43.4	42.9	48.5	53.7	56.3	62.9	61.2	56.5	53.0	49.7	46.0
	2%	DB	45.6	44.9	45.5	51.4	57.5	60.3	66.7	65.2	60.9	56.0	51.8	47.8
		MCWB	43.0	42.4	42.5	45.7	51.4	54.2	60.8	59.6	55.4	51.2	48.6	44.9
	5%	DB	44.6	43.8	44.5	48.4	54.7	58.0	63.9	63.3	59.6	54.7	50.4	46.6
		MCWB	42.0	41.3	41.9	44.1	50.0	53.5	59.0	58.4	54.8	50.4	46.9	43.5
	10%	DB	43.3	42.8	43.5	46.2	52.3	56.1	61.5	61.6	58.4	53.1	48.8	45.3
		MCWB	40.7	40.3	41.0	42.8	48.5	52.5	57.7	57.4	54.2	49.3	45.4	42.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.4	44.5	44.8	48.8	54.5	57.7	64.2	61.8	57.6	53.9	50.9	47.5
		MCDB	46.4	45.6	45.6	53.8	59.9	62.2	69.6	66.4	60.1	56.5	52.5	48.4
	2%	WB	43.9	43.0	43.7	46.6	52.2	55.2	61.6	60.3	56.7	52.1	49.1	45.6
		MCDB	45.3	44.1	45.0	50.8	56.5	58.5	65.9	64.1	59.2	55.1	51.3	47.2
	5%	WB	42.3	41.7	42.3	44.8	50.5	53.8	59.8	59.1	55.9	50.9	47.5	43.9
		MCDB	44.2	43.4	44.1	47.4	54.2	57.0	63.3	62.6	58.6	54.0	50.1	46.0
	10%	WB	40.8	40.6	41.2	43.5	48.7	52.8	58.2	57.8	55.1	49.7	45.8	42.5
		MCDB	43.1	42.6	43.3	45.7	51.9	55.6	61.5	60.9	57.9	52.9	48.7	45.0
Mean Daily Temperature Range	5% DB	MDBR	5.1	4.9	5.0	5.3	5.9	5.3	5.6	5.1	4.7	4.4	4.8	5.1
		MCDBR	5.4	4.7	5.2	7.3	9.6	8.6	9.5	7.6	5.8	5.1	5.0	6.2
		MCWBR	5.7	5.2	5.0	4.7	5.6	5.4	5.6	4.5	4.1	4.6	5.0	6.7
	5% WB	MCDBR	5.3	4.7	5.1	6.3	9.2	7.7	8.8	7.1	5.4	4.8	5.0	6.2
		MCWBR	5.8	5.4	5.3	4.5	5.8	5.3	5.6	4.6	4.2	4.7	5.5	6.7
Clear-Sky Solar Irradiance	taub	0.382	0.317	0.340	0.359	0.353	0.356	0.363	0.352	0.341	0.315	0.353	0.360	
	taud	2.564	2.242	2.326	2.375	2.458	2.485	2.491	2.517	2.515	2.388	2.517	2.446	
	Ebn at Noon	94	186	236	259	272	274	268	261	239	189	93	38	
	Edh at Noon	9	18	25	30	30	30	29	26	21	16	8	5	
All-Sky Solar Radiation	RadAvg	36	176	553	1097	1535	1630	1500	1153	677	283	62	15	
	RadStd	5	27	76	96	73	164	141	108	83	42	11	1	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (26 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page